Total no. of Pages: 1 Roll no. ....

Degree: B-Tech. Semester 3rd

MID-SEMESTER EXAMINATION (summer), June, 2026

Course Title: Probability Theory and Random Process
Course Code: Sem: ECECC06

Duration: 1:30 Hours Max. Marks: 15

Note: - Attempt all questions in the given order only. Missing data/information (if any), may be suitably assumed & mentioned in the answer.

<table>
<tr><th>Q. No.</th><th>Question</th><th>Marks</th><th>CO</th></tr>
<tr><td>1a</td><td>Define a Random Variable. Differentiate between discrete and continuous random variables.</td><td>1.5</td><td>C01</td></tr>
<tr><td>1b</td><td>Let X be a continuous random variable with the following distribution:
$$f(x) = \begin{cases} kx, & if\ 0 \le x \le 5 \\ 0, & elsewhere \end{cases}$$
(a) Evaluate k. (b) Find $P(1 \le X \le 3)$.</td><td>1.5</td><td>C02</td></tr>
<tr><td>2a</td><td>Distinguish between independent and uncorrelated random variables.</td><td>1.5</td><td>C01</td></tr>
<tr><td>2b</td><td>A fair coin is tossed four times. Let X denote the number of heads occurring. Find: (a) the distribution of X, (b) E[X], (c) the probability graph of X.</td><td>1.5</td><td>C02</td></tr>
<tr><td>3a</td><td>If E[X] = 3, E[Y]=2 and E[XY]= 8, $Cov(X,Y)$.</td><td>1.5</td><td>C02</td></tr>
<tr><td>3b</td><td>Let X and Y be random variables with join distribution given below:
<table>
<tr><th>X \ Y</th><th>-3</th><th>2</th><th>4</th><th>Sum</th></tr>
<tr><td>1</td><td>0.1</td><td>0.2</td><td>0.2</td><td>0.5</td></tr>
<tr><td>3</td><td>0.3</td><td>0.1</td><td>0.1</td><td>0.5</td></tr>
<tr><td>Sum</td><td>0.4</td><td>0.3</td><td>0.3</td><td></td></tr>
</table>
Find: (a) the distributions of X and Y.</td><td>1.5</td><td>C02</td></tr>
<tr><td>4a</td><td>If $X \sim N(\mu, \sigma^2)$ with $\mu = 20$ and $\sigma = 4$, write the expression for its PDF.</td><td>1.5</td><td>C02</td></tr>
<tr><td>4b</td><td>A fair coin is tossed 100 times. Find the probability P that the heads occur: (a) exactly 60 times, (b) between 48 and 53 times exclusive (c) less than 45 times.</td><td>1.5</td><td>C02</td></tr>
<tr><td>5a</td><td>Prove $Var(aX + b) = a^2 Var(X)$</td><td>1.5</td><td>C01</td></tr>
<tr><td>5b</td><td>Let X be the binomial random variable B(n, p). Show that (i) $\mu = E[x] = np$ (ii) Var(X) = npq.</td><td>1.5</td><td>C02</td></tr>
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